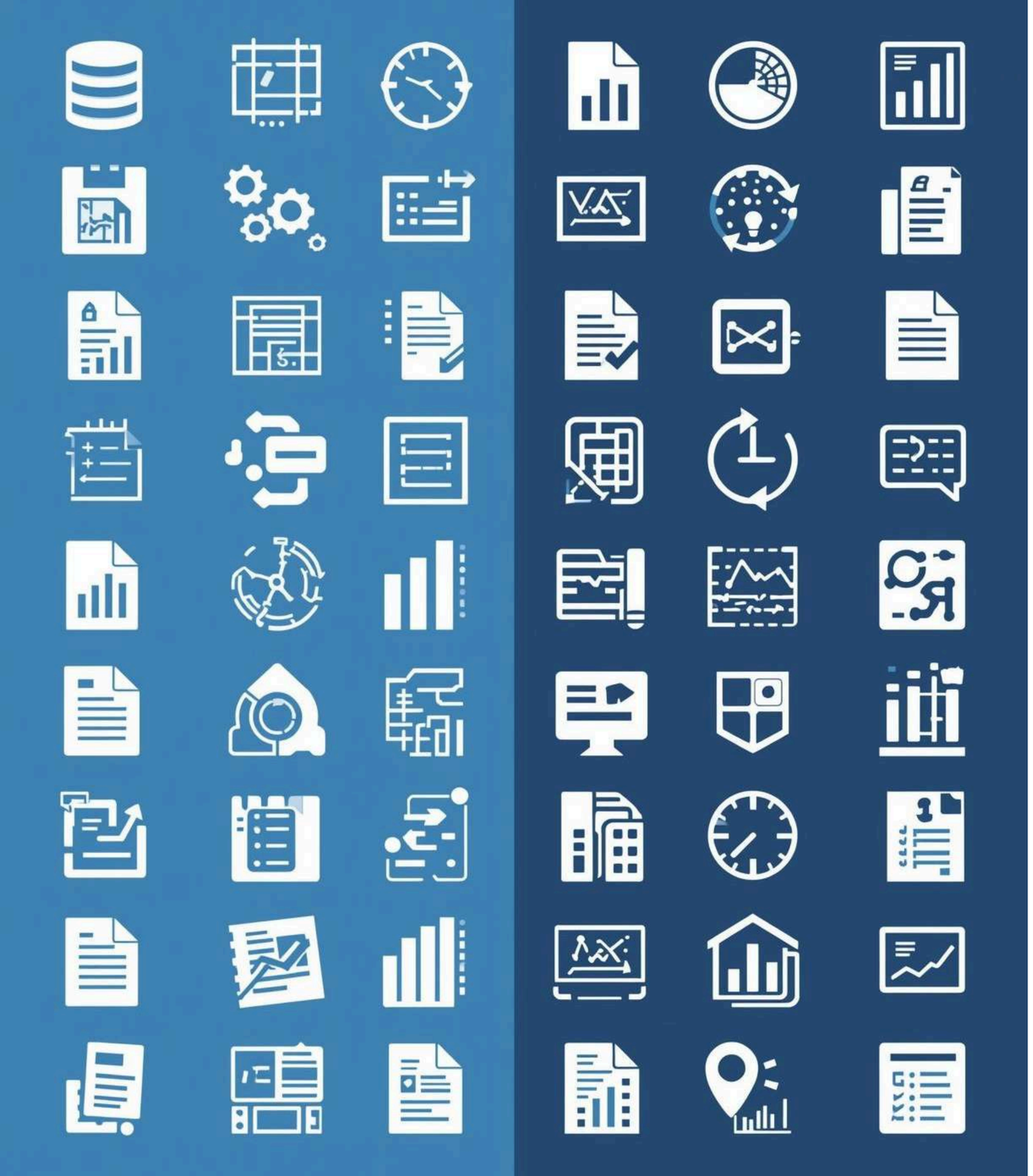


Session 1

Learning Aim B: Data vs Information

Introduction

Today we are starting **Learning Aim B**, which is about **data and information in organisations.**

In the real world, **every organisation uses data every day.**
Schools, hospitals, companies, shops, banks — all of them collect data and turn it into information.

But before we talk about systems, reports, or decisions,
we must first understand **one very important question:**

What is the difference between data and information?

If you understand this difference well,
everything else in Learning Aim B will become much easier.

What is Data?

Data is raw facts.

Data by itself **does not explain anything.**

Examples of data:

- Numbers: 25, 40, 100
- Names: Ali, Sara
- Dates: 12/11/2024
- Text: “Delivered”, “Pending”

Data is **unprocessed**.

It has no meaning unless we add context.

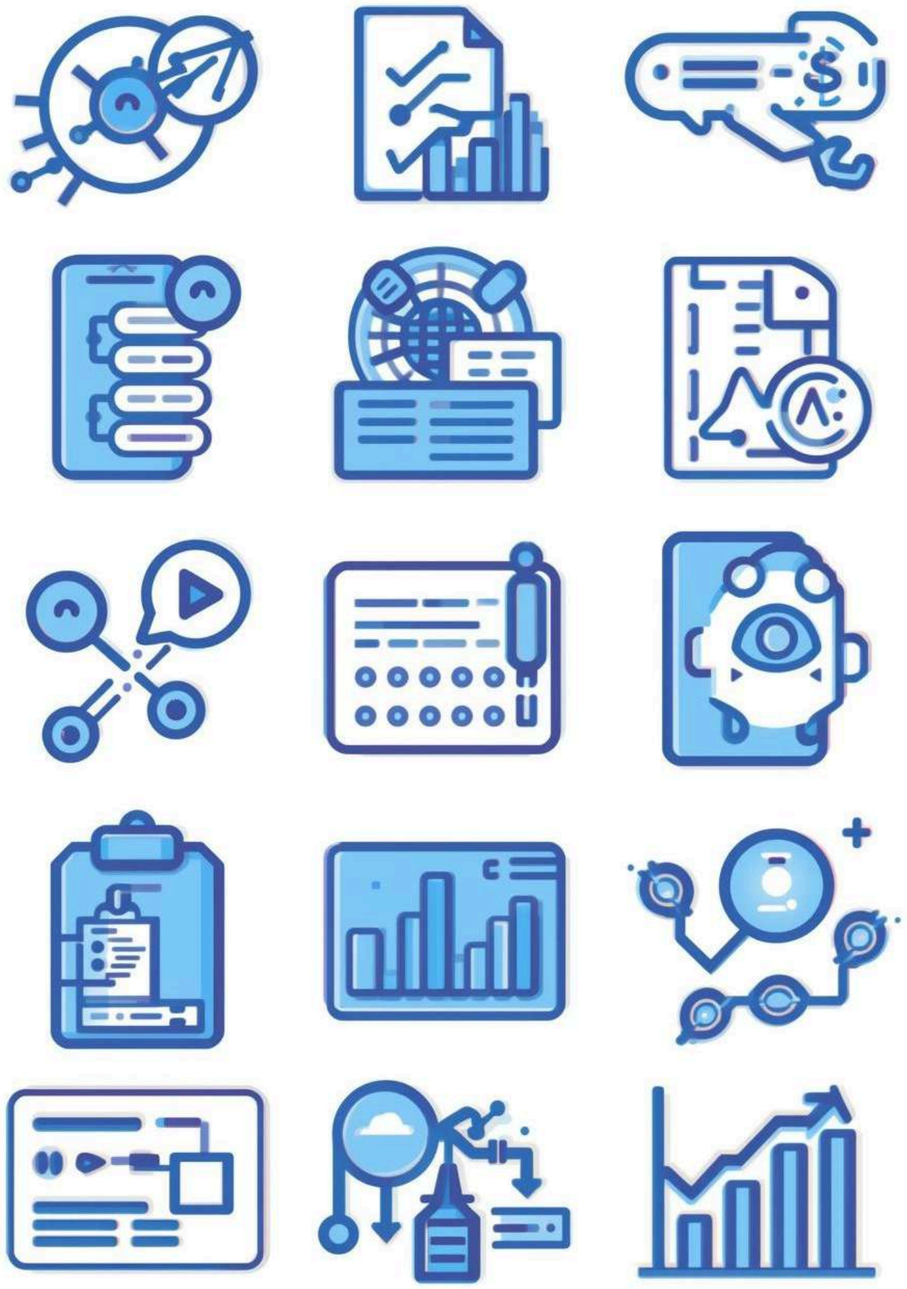
For example:

- “50”
- This is data.
- We do not know what it means.

Is it:

- 50 customers?
- 50 degrees?
- 50 products?

We cannot make decisions using data alone.



What is Information?

Information is data that has been:

- organised
- processed
- analysed
- given meaning

Information helps organisations **understand what is happening.**

Example:

- Data:
- 50, 60, 55
- Information:
- “The number of customers increased this week.”

Now the organisation can:

- understand the situation
- make decisions
- plan actions

So we can say:

Information = Data + Meaning + Context

Data vs Information

DATA: RAW FACTS

Data is simple facts and numbers.
By itself, data has no clear meaning.
It cannot help with decisions unless it is processed.
Example:
50, 2024, Ali

INFORMATION: MEANINGFUL

Information is data that has been organized and explained.
It has meaning and helps organizations understand situations and make decisions.
Example:
“50 students attended the class in 2024.”

Why Organisations Cannot Use Data Alone

Organisations do not make decisions using raw data.

Imagine a hospital:

- Data: heart rate numbers, temperature readings
- Information: patient condition report

Doctors cannot treat a patient by looking at numbers only.

They need information, not just data.

The same idea applies to:

- schools
- businesses
- logistics companies
- online platforms

Data vs Information – Simple Comparison

Data	Information
Raw facts	Processed data
No meaning alone	Has meaning
Hard to use	Used for decisions
Example: “120”	“120 orders today”

This difference is very important in IT systems.

Understanding Organisational Levels of Information

HOW DIFFERENT LEVELS UTILIZE DATA EFFECTIVELY

Organisations operate at three levels: **operational**, **management**, and **strategic**. Each level uses information differently to support decision-making, ensuring efficient functioning and long-term success.

1- Operational Level

This is the day-to-day level.

Used by:

- employees
- staff
- technicians

They need real-time data.

Examples:

- stock levels
- number of customers today
- available resources
- daily schedules

Example:

In a retail shop:

- Data: number of items sold today
- Information: “Only 5 items left in stock”

Management Level

This level is used by managers.
They need summarised information, not raw data.
Examples:

- monthly sales reports
- employee performance
- customer complaints
- financial summaries

Example:
In a hospital:

- Information: “Patient waiting time increased by 20% this month”

This helps managers improve services.

Strategic Level

This is the highest level in an organisation.

Used by:

- directors
- owners
- senior management

They focus on:

- long-term decisions
- future planning
- growth and change

Examples:

- opening a new branch
- investing in new technology
- entering new markets

Example:

A company uses information to decide:

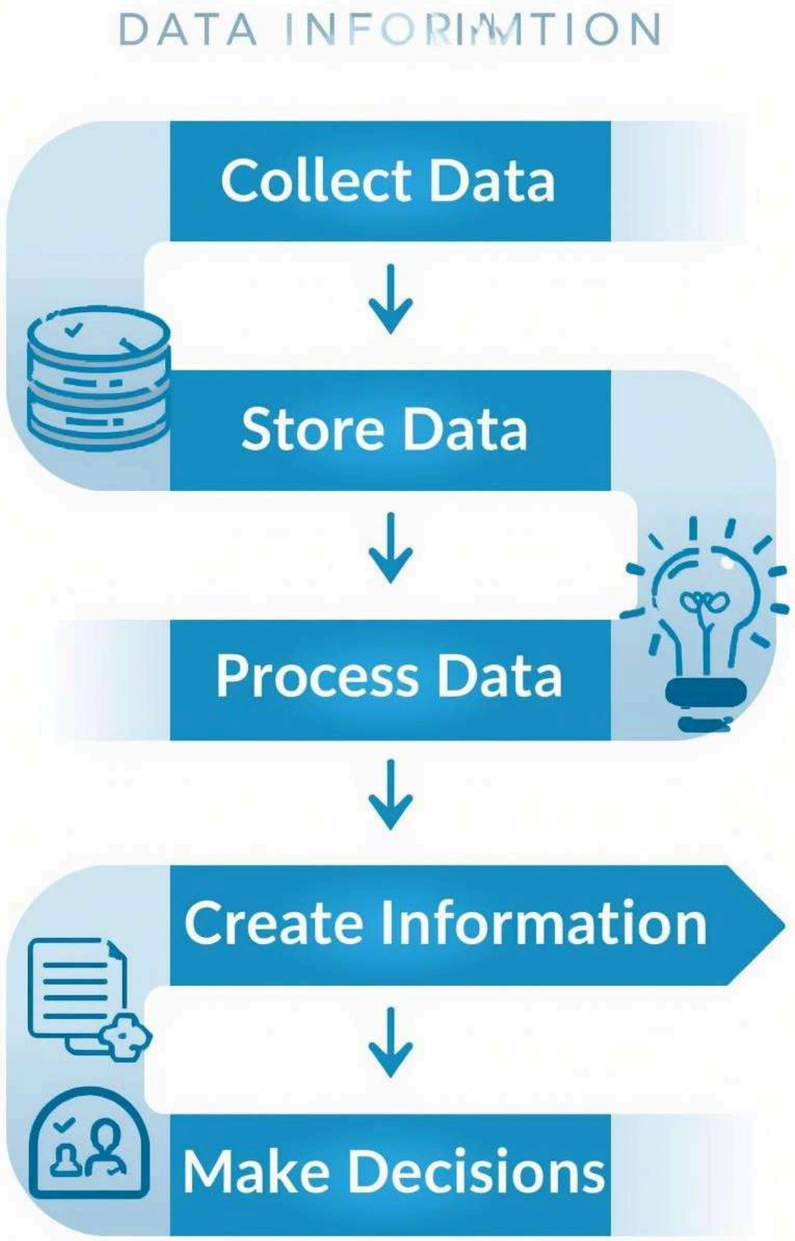
- Should we expand?
- Should we change our services?

How Data Becomes Information in an Organisation

The process is usually:

1. Data is collected
2. Data is stored
3. Data is processed
4. Information is created
5. Decisions are made

IT systems help at every step of this process.

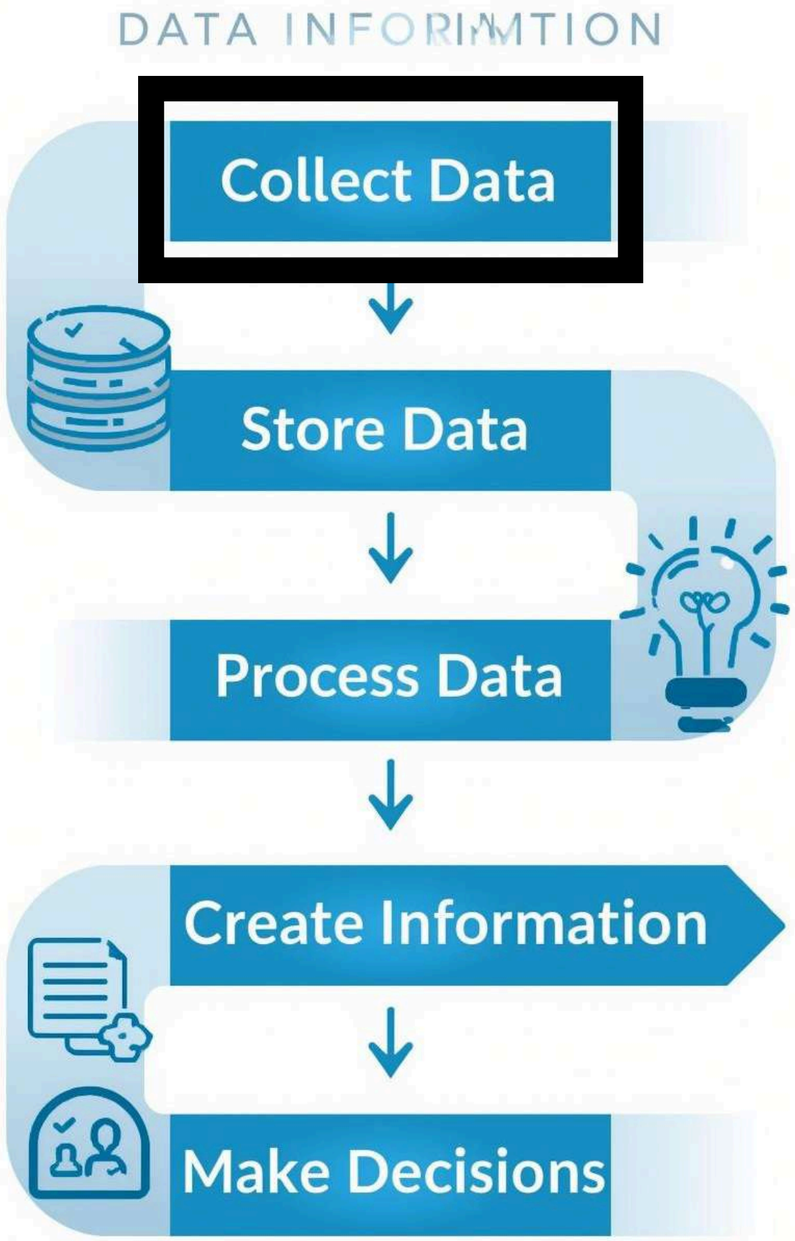


1) Data is collected

Data is raw facts. It has no meaning by itself.
Organisations collect data from many sources, such as:

- online forms (Google Forms)
- POS systems in shops (Point of Sale)
- sensors (temperature, heart rate, motion)
- attendance systems
- customer transactions

Example (Retail shop):
Every sale creates data: product name, price, quantity, date, time.
Example (Hospital):
Patient data: temperature, blood pressure, test results.



2) Data is stored

After collecting data, organisations must store it safely so they can use it later.

Storage can be:

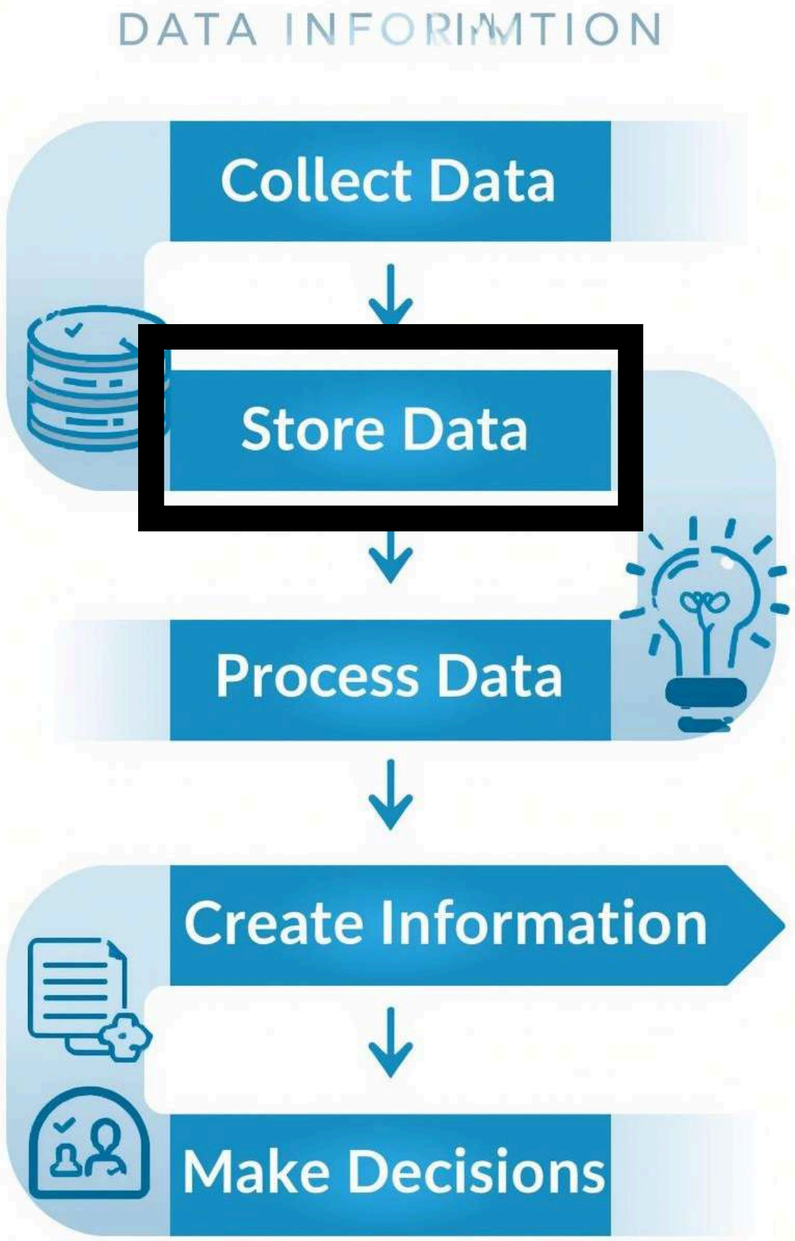
- databases (SQL)
- Excel sheets (small organisations)
- cloud storage (Google Drive / Microsoft 365)
- special systems (hospital system, student system)

Example (Shop):

Sales data is stored in a database so the shop can check sales and stock later.

Example (University):

Student grades and attendance are stored in a system to track progress.



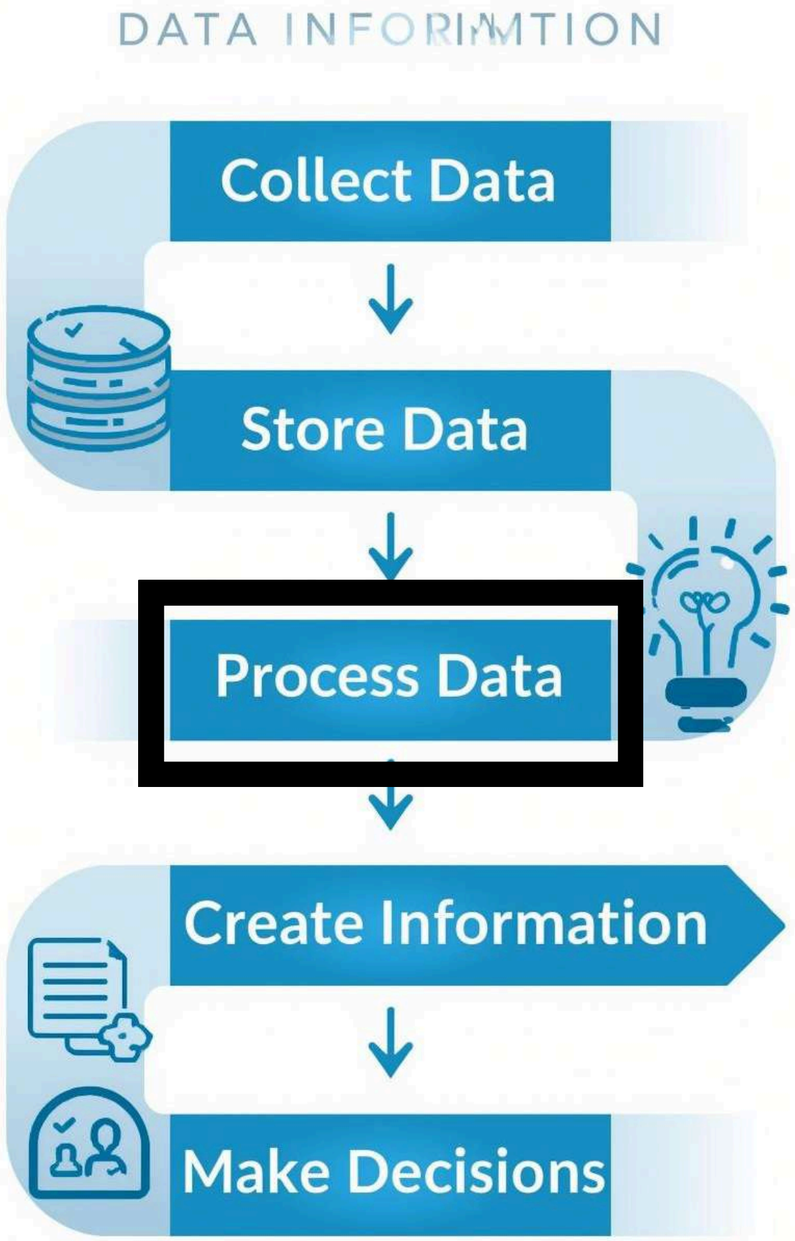
3) Data is processed

Processing means the system works on the data to make it useful, such as:

- sorting (by date, product, branch)
- filtering (show only one month or one department)
- calculating (total, average, percentage)
- grouping (by category, week, customer type)

Example (Shop):
The system calculates total sales for each product in one week.

Example (Hospital):
The system compares a patient's readings to normal ranges and finds issues.



4) Information is created

Information = processed data with meaning and context.

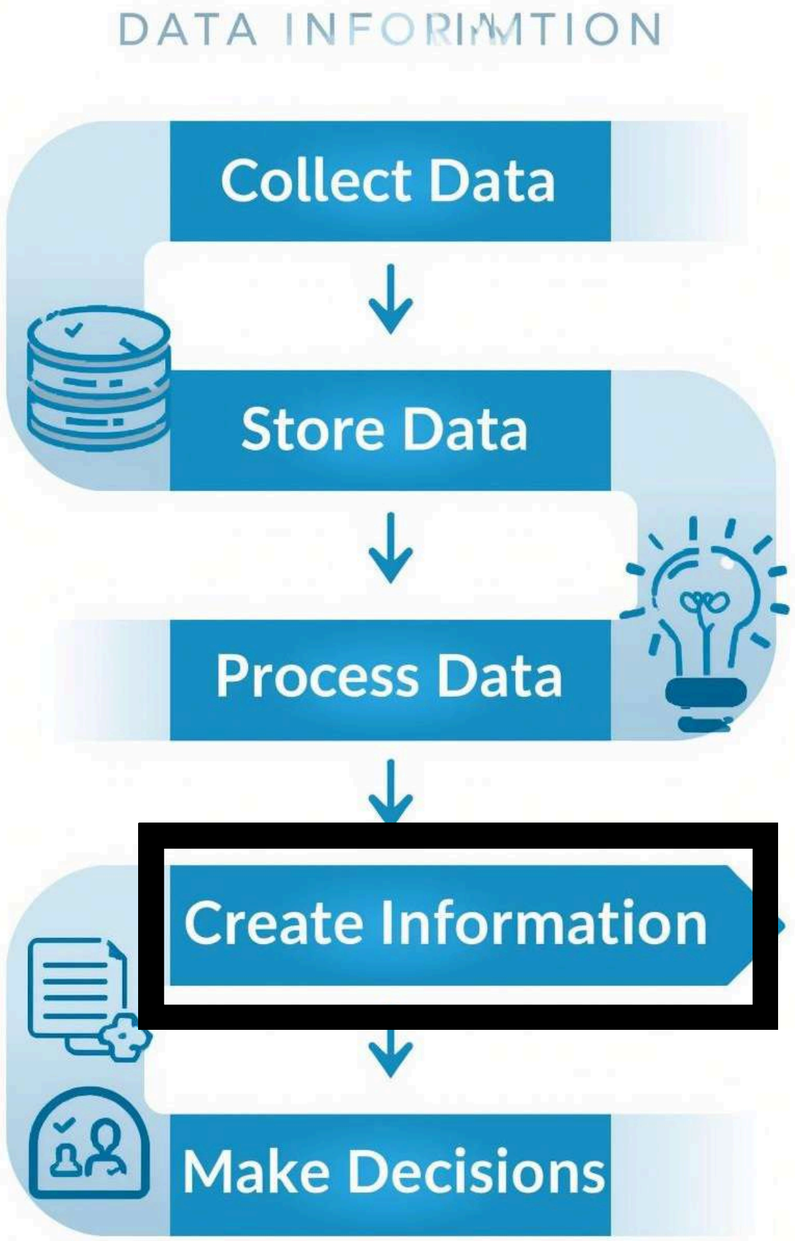
This is the point where we can understand what is happening.

Example (Shop):

- “Product X is almost out of stock (only 5 left).”
- “Sales increased by 20% this week.”

Example (University):

- “This student’s attendance is below 75%.”
- “The class average dropped in Quiz 2.”



5) Decisions are made

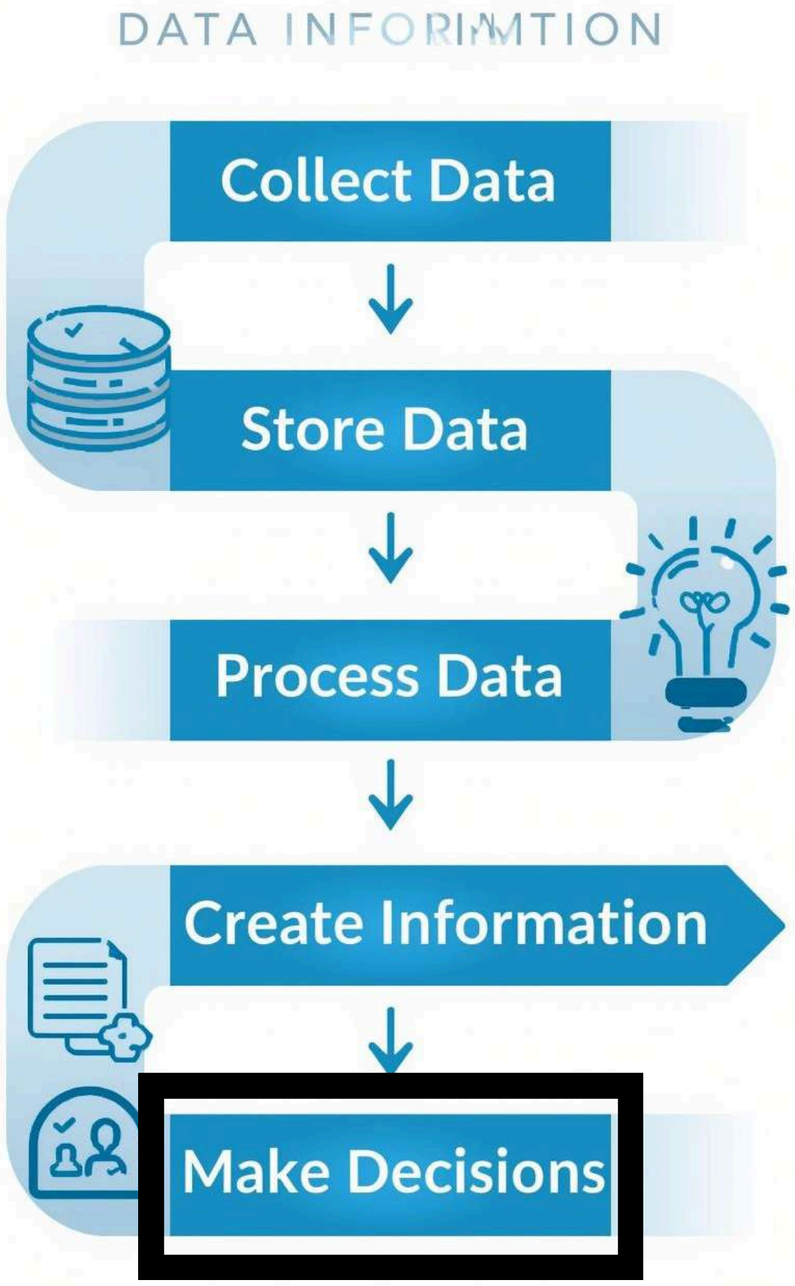
Finally, organisations use information to make decisions.

Example (Shop):

- Order more stock
- Create discounts for slow products
- Add more staff during busy days

Example (Hospital):

- Give urgent treatment
- Change medication dose
- Schedule more tests



How IT systems help at every step

IT systems make this process:

- faster
- more accurate
- easier to manage
- less error-prone

They help with:

- collecting data automatically (forms, sensors, POS)
- storing data securely (databases, cloud, backups)
- processing data quickly (reports, calculations)
- presenting information clearly (dashboards, charts)
- supporting decisions (alerts,, summaries)